

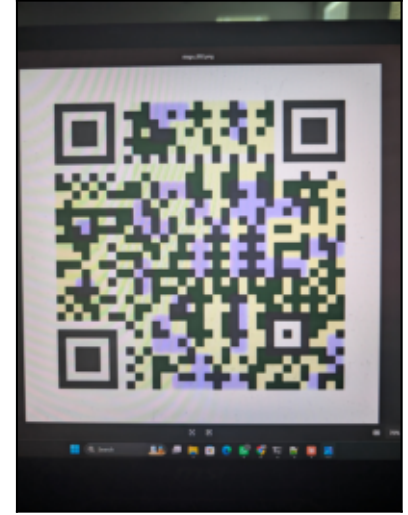
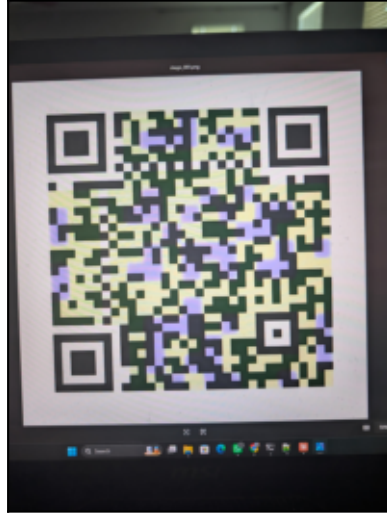
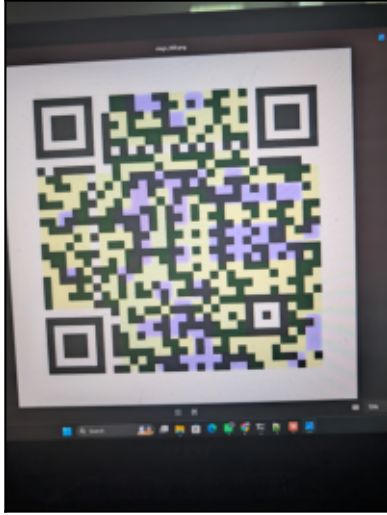
EXP-002 — stego QR displayed on a monitor, photographed with a phone

PXL_20260615_135555817.jpg

PXL_20260615_135605657.jpg

PXL_20260615_135615592.jpg

raw phone photos



OpenCV localize + perspective-rectify
→ decoder input

Representative decode

```
public payload (pyzbar):      '8FG4N24JW1QG'  
hidden message (neural+ECC): d08f457c80 ✓ matches ground truth  
decoder confidence:          0.731
```

Aggregate over all 10 photos

```
QR located:                   10 / 10  
public decode (pyzbar):      10 / 10  
hidden MESSAGE decode:      10 / 10
```

*100% recovery through a real display → camera channel
(perspective · glare · moiré · JPEG · screen colour),
from a model trained only on the simulated distortion layer.*

